

WaveLock: A Curvature-Locked One-Way Function Based on Nonlinear PDE Evolution

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Abstract

This paper introduces *WaveLock*, a new empirical family of **curvature-locked one-way functions (CLOWF)** constructed from deterministic nonlinear partial differential equation (PDE) evolution on a compact two-dimensional lattice. A WaveLock instance begins with an initial wavefield ψ_0 , evolves it under a fixed nonlinear curvature operator to obtain a terminal field ψ^* , and derives a cryptographic commitment $C = H(\text{Serialize}(\psi^*))$ using SHA-256. WaveLock is not proposed as a replacement for classical hash functions. Instead, it defines a new design space where *cryptographic one-wayness* may emerge from geometric, curvature-based irreversibility. We evaluate its empirical properties using an extensive adversarial test suite comprising gradient-based inversion, Jacobian backpropagation, tangent-space spectral analysis, reverse-time orbit construction, neural inverse mapping, Fourier and wavelet reconstruction, Monte-Carlo sampling, adjoint-PDE inversion, and several quantum-inspired simulated attacks. Across all tested adversaries, no reconstruction of ψ^* or collision with C was observed. We do not claim formal cryptographic security; WaveLock is presented as a **research artifact** motivating further study of curvature-based one-way functions.

1 Introduction

Classical cryptographic hash functions such as SHA-2 [9], SHA-3 [10], and BLAKE3 [12] derive their pseudorandomness and one-wayness from bitwise permutations, Boolean circuits, and algebraic diffusion. This paper explores a complementary direction: constructing cryptographic one-way behavior using *nonlinear PDE evolution*.

The core observation motivating WaveLock is that contractive nonlinear dynamical systems destroy information irreversibly. A forward evaluation is efficient—it requires only T applications of a local lattice operator—while inversion demands recovering an exponentially large preimage set. This asymmetry is structural rather than algebraic: it arises from the geometry of the evolution operator, not from number-theoretic hardness assumptions.

WaveLock is an empirical construction combining four ingredients: deterministic curvature evolution on a periodic lattice, strong FP64 sensitivity to perturbations, nonlinear reaction–diffusion–feedback dynamics, and structured serialization prior to hashing. We evaluate whether such a system can exhibit cryptographically relevant properties—determinism, avalanche behavior, resistance to

structured inversion attacks, and resistance to quantum-inspired heuristic attacks—without claiming that WaveLock is a provably secure cryptographic primitive.

2 Related Work

Classical hash functions. The SHA-2 family [9] and SHA-3 (Keccak) [10] achieve one-wayness through iterated algebraic mixing of fixed-width state vectors. BLAKE3 [12] adds a Merkle tree structure for parallelism. All three derive security from the avalanche properties of their round functions, which are designed by hand and evaluated through extensive cryptanalysis.

Physics-inspired cryptographic primitives. The idea of basing cryptographic hardness on physical processes has appeared in several forms. Verifiable delay functions (VDFs) [4] exploit the sequential nature of repeated squaring, which is algebraic but conceptually related to iterated evolution. Optical and photonic physical unclonable functions (PUFs) [13] derive one-wayness from the complexity of light scattering in a disordered medium. WaveLock differs from PUFs in being fully deterministic and software-defined, and from VDFs in using nonlinear PDE contraction rather than algebraic sequential computation.

Nonlinear dynamics and chaos in cryptography. Chaotic maps have been used extensively in image encryption and pseudorandom number generation [2, 6]. However, chaotic systems are typically *invertible* (e.g., the Lorenz or Hénon maps are diffeomorphisms), and their cryptographic security often relies on sensitivity to initial conditions rather than on information destruction. WaveLock’s operator is a strict contraction, making it fundamentally non-invertible—a qualitative difference from chaos-based schemes.

Reaction–diffusion systems. The nonlinear operator in WaveLock draws on the structure of reaction–diffusion equations [8], with a curvature-feedback term inspired by the Θ -operator of Wave Confinement Theory [15]. The Allen–Cahn equation and Cahn–Hilliard equation are well-studied models of phase separation with contractive long-time behavior [1], but they have not previously been proposed as cryptographic primitives.

Post-quantum cryptography. Lattice-based [14], code-based [7], and hash-based [3] schemes form the basis of NIST’s post-quantum standardization effort [11]. WaveLock’s security does not rest on any of these algebraic structures; instead, it offers a complementary direction in which hardness arises from the contraction and information loss inherent in nonlinear PDE dynamics.

3 WaveLock Construction

3.1 Initial Wavefield

Let $\psi_0 \in \mathbb{R}^{N \times N}$ be a double-precision (FP64) floating-point grid with $N = 2^{n/2}$. In all experiments reported here, $n = 6$ and $N = 32$, giving $d = N^2 = 1024$ degrees of freedom. The field ψ_0 is generated deterministically from an integer seed s using a seeded pseudorandom generator.

3.2 Evolution Operator

WaveLock evolves ψ via forward Euler integration:

$$\psi_{t+1} = \psi_t + \Delta t \cdot F(\psi_t), \quad (1)$$

where $\Delta t = 0.1$ is the fixed timestep and $T = 50$ is the total number of iterations. The forcing operator is

$$F(\psi) = \underbrace{\frac{\alpha \Delta \psi}{\psi + \varepsilon \exp(-\beta \psi^2)}}_{\text{curvature feedback}} - \underbrace{\theta \psi \Delta \log(\psi^2 + \delta)}_{\text{entropy diffusion}} - \underbrace{\mu \psi}_{\text{damping}}, \quad (2)$$

with the following fixed parameters:

Table 1: WaveLock evolution parameters.

Symbol	Value	Role
α	1.50	Curvature feedback strength
β	2.6×10^{-3}	Gaussian suppression width
θ	1.0×10^{-5}	Entropy diffusion coefficient
ε	1.0×10^{-12}	Denominator regularizer
δ	1.0×10^{-12}	Logarithm regularizer
μ	2.0×10^{-5}	Linear damping rate
Δt	0.1	Forward Euler timestep
T	50	Number of evolution steps

Here Δ denotes the discrete Laplacian computed with a standard five-point stencil on a periodic (toroidal) grid:

$$(\Delta \psi)_{i,j} = \psi_{i+1,j} + \psi_{i-1,j} + \psi_{i,j+1} + \psi_{i,j-1} - 4\psi_{i,j}, \quad (3)$$

with indices taken modulo N .

The curvature feedback term is the discrete analogue of the nonlinear curvature operator $\Theta[\psi] = -\nabla^2 \psi / (\psi + \varepsilon e^{-\beta|\psi|^2})$ introduced in Wave Confinement Theory [15]. The entropy diffusion term introduces a second-order logarithmic nonlinearity that breaks spectral regularity. Together with the linear damping term, these three components produce a strongly contractive, information-destroying evolution.

The complete evolution procedure is given in Algorithm 1.

3.3 Terminal Field

After T steps the system yields a terminal field $\psi^* = \Phi_T(\psi_0)$, which is deterministic given ψ_0 . As we show in Section 4, perturbations of magnitude 10^{-6} to either ψ^* or ψ_0 produce globally divergent outputs.

3.4 Commitment Function

WaveLock computes a cryptographic commitment:

$$C = H(\text{Serialize}(\psi^*)), \quad (4)$$

Algorithm 1 WaveLock PDE evolution: $\psi_0 \mapsto \psi^*$.

Require: Seed s , grid size $N = 2^{n/2}$, parameters as in Table 1

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1:  $\psi \leftarrow \text{InitField}(s, N)$  {deterministic FP64 field}
2: for  $t = 1, \dots, T$  do
3:    $L \leftarrow \Delta\psi$  {5-point Laplacian, periodic BC}
4:    $\text{fb} \leftarrow \alpha \cdot L / (\psi + \varepsilon \cdot \exp(-\beta \psi^2))$ 
5:    $\text{ent} \leftarrow \theta \cdot \psi \cdot \Delta \log(\psi^2 + \delta)$ 
6:    $\psi \leftarrow \psi + \Delta t \cdot (\text{fb} - \text{ent}) - \mu \cdot \psi$ 
7: end for
8: return  $\psi^* \leftarrow \psi$ 

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where Serialize encodes the FP64 matrix in a canonical byte order together with a JSON header containing the evolution parameters, schema version, and a kernel integrity hash. The hash H is standard SHA-256 [9]. WaveLock does not aim to replace SHA-256; its novelty lies in the difficulty of reconstructing ψ^* from C .

3.5 Computational Cost

A single WaveLock evaluation consists of $T = 50$ applications of the forcing operator (2) on an $N \times N$ grid. Each step requires two Laplacian evaluations (one for the feedback term, one inside the entropy term), one element-wise exponential, one element-wise logarithm, and $O(N^2)$ arithmetic. The total cost is therefore $O(TN^2)$, which for the default parameters ($T = 50$, $N = 32$) amounts to approximately 1.6×10^5 floating-point operations plus one SHA-256 evaluation. On a modern GPU the forward pass completes in under one millisecond.

4 Empirical Properties

4.1 Determinism

Replicating evolution with identical inputs (same seed, same parameters, same FP64 arithmetic) produces identical ψ^* , identical serialized payloads, and identical SHA-256 commitments with zero Hamming-distance deviation.

4.2 Avalanche Sensitivity

Perturbing a single cell of ψ^* by 10^{-4} produces a fully divergent commitment: the resulting SHA-256 digest differs in approximately half of its bits, demonstrating empirical avalanche behavior.

4.3 Collision Search Resistance

Across hundreds of seeds and thousands of perturbed fields, no commitment collisions were observed. A parallelized collision search over the seed range $\{0, 1, \dots, 10^5\}$ produced 100,000 distinct SHA-256 digests with no duplicates. This is an empirical observation, not a proof of collision resistance.

Table 2: Summary of adversarial attacks on WaveLock. No attack achieved reconstruction of ψ^* or ψ_0 , nor produced a commitment collision.

Attack class	Method	Effort	Result
Classical inversion	Gradient surrogate	3000 iters	No convergence
	Fourier reconstruction	Full spectrum	No match
	Wavelet (Haar) multiscale	Scales 1–16	No match
	Adjoint PDE inversion	Reverse-time	No convergence
	Random projection	10^4 trials	No match
Stochastic	Monte-Carlo annealing	Boltzmann sched.	No match
	Thermal noise	Gaussian fields	No match
White-box	TBJA (full backprop)	2000 iters/GPU	Loss floor ≈ 0.08
	Jacobian spectrum	Power iteration	$\sigma_{\max} \approx 0.50$
	Lyapunov–Perron reverse	1500 iters	Endpoint loss $\approx 5 \times 10^3$
	Neural inverse (CNN)	20 epochs	Test loss ≈ 4.5
Quantum-inspired	Grover-Sim	Amplitude amp.	No match
	QAOA-Sim	Variational opt.	No match
	QFT reconstruction	Spectral	No match
	Gibbs QSim	QMC sampling	No match
	Quantum random walk	Unitary diff.	No match

5 Adversarial Cryptanalysis

This section reports empirical results from a comprehensive suite of inversion and reconstruction attacks. Table 2 summarizes all attacks and their outcomes; detailed descriptions follow.

5.1 Classical Inversion Attacks

Gradient surrogate attack. A Laplacian-based gradient descent was applied to minimize $\|\Phi_T(\hat{\psi}_0) - \psi^*\|^2$. No convergence was observed within 3000 iterations.

Fourier reconstruction. Both low-frequency and full-spectrum Fourier coefficient matching were attempted. No reconstruction matched the target commitment.

Wavelet multiscale reconstruction. Haar-basis reconstruction at scales 1 through 16 was attempted. No match was observed.

Adjoint PDE inversion. Reverse-time evolution using the adjoint of the linearized WaveLock operator failed to converge to ψ^* .

Random projection. Linear combinations of ψ^* with random noise fields were evaluated. No accidental commitment match was found across 10^4 trials.

5.2 Stochastic Attacks

Monte-Carlo annealing. Boltzmann-style simulated annealing across multiple temperature schedules produced no commitment match.

Thermal noise perturbation. Perturbation of ψ^* by i.i.d. Gaussian noise fields of varying magnitude produced no accidental matches.

5.3 White-Box Attacks

True Backpropagation Jacobian Attack (TBJA). We implemented a differentiable surrogate of the WaveLock evolution operator and computed exact gradients $\nabla_{\psi_0} \psi^*(\psi_0)$ via end-to-end reverse-mode differentiation through all $T = 50$ PDE steps. Adam optimization was applied over 2000 iterations on GPU. For $n = 6$ (32×32 grid) the attack achieved a best loss of approximately 8.15×10^{-2} and stagnated at a nonzero floor, indicating that the forward map is not invertible even under exact gradient information.

Tangent-space collapse (Jacobian spectrum analysis). The top singular value of the end-to-end Jacobian $J = \partial \psi^* / \partial \psi_0$ was estimated via Hutchinson-style power iteration. For $n = 4$ we obtained $\sigma_{\max}(J) \approx 0.50$, implying $\|J\|_2 < 1$ and confirming that the WaveLock PDE is a contraction mapping. Over T iterations the effective spectral radius decays as $\sigma_{\max}^T \approx (0.5)^{50} \approx 8.9 \times 10^{-16}$, demonstrating exponential collapse of local perturbations.

Lyapunov–Perron reverse-time synchronization. We attempted to construct a reverse-time trajectory $\psi_0, \psi_1, \dots, \psi_T = \psi^*$ by jointly optimizing all intermediate fields under the constraints $\psi_{t+1} \approx F(\psi_t)$ and $\psi_T \approx \psi^*$. After 1500 optimization steps the total loss remained at approximately 5.28×10^4 with an endpoint loss of approximately 5.28×10^3 . Forward consistency improved, but endpoint constraints remained far from satisfied, strongly indicating that no backward orbit consistent with the PDE terminates at ψ^* .

Neural inversion attack. A small convolutional neural network f_θ was trained to approximate $f_\theta(\psi^*) \approx \psi_0$ using hundreds of WaveLock-generated training pairs. After 20 epochs the training loss was approximately 1.88 and the test loss approximately 4.52. The inverse map is not learnable: even coarse statistical structure cannot be inverted to meaningful reconstructions.

5.4 Quantum-Inspired Attacks

Five quantum-inspired classical simulations were evaluated: simulated Grover amplitude amplification, QAOA-style variational optimization, QFT-based spectral reconstruction, Gibbs quantum Monte-Carlo sampling, and quantum random walk diffusion through ψ -space. None produced a reconstruction of ψ^* or ψ_0 , nor a commitment collision.

6 Empirical Irreversibility and Information Collapse

The adversarial evaluation reveals a coherent picture of WaveLock as an empirically irreversible dynamical system.

6.1 Strong Contraction of the Forward Map

The empirically measured Jacobian spectral bound $\sigma_{\max} \approx 0.5$ implies $\|J\|_2 < 1$, so the single-step map Φ is contractive. After T iterations, perturbation magnitudes decay as $(0.5)^T \approx 10^{-15}$.

6.2 Many-to-One Behavior

Both the TBJA and the Lyapunov–Perron inversion demonstrate that no gradient path or reverse orbit maps ψ^* back to its originating ψ_0 . The PDE evolution satisfies $\Phi_T^{-1}(\psi^*) = \emptyset$ in the functional sense: no single-valued inverse exists.

6.3 Collision Search for the Composite Map

In addition to analyzing the invertibility of Φ_T , we evaluated whether the composite commitment map $G = H \circ \text{Serialize} \circ \Phi_T$ exhibits accidental structural collisions. A parallelized search over 100,000 seeds with four workers found no collisions: every seed produced a unique SHA-256 digest. This does not constitute a cryptographic proof of collision resistance, but combined with the contraction properties of Φ_T it provides evidence that no low-complexity structural collisions arise from the PDE evolution.

Remark 6.1. Since SHA-256 is collision-resistant under standard assumptions [9], and Φ_T destroys geometric and spectral structure through contraction, the composite map G inherits no obvious degeneracies that would facilitate collision search within the tested domain.

6.4 Inverse Map Not Learnable

Neural inversion experiments show that ψ^* retains insufficient structure to reconstruct ψ_0 even approximately. Test losses remained on the order of 1–10, confirming that the inverse problem is not statistically learnable.

6.5 Collapse of Local and Global Structure

The combination of contraction, large endpoint losses, and failed manifold approximations indicates that WaveLock evolution collapses both local tangent-space structure and global manifold structure of initial conditions. The PDE evolution behaves as a nonlinear attractor with significant dimensional reduction.

6.6 Implications for One-Way Behavior

WaveLock is not proposed as a cryptographically secure primitive. However, the observed behavior $\psi_0 \rightarrow \psi^* \rightarrow H(\text{Serialize}(\psi^*))$ is empirically one-way because: (i) Φ is contractive and information-destroying, (ii) Φ is non-injective, (iii) Φ^{-1} does not exist, (iv) the inverse map is not learnable, and (v) serialization followed by hashing further eliminates recoverable structure. These properties motivate further analysis of curvature-based one-way functions constructed from nonlinear PDE evolution.

7 Discussion

WaveLock displays empirical resistance to a broad range of tested attacks. However, these results do not constitute a security proof. Several important caveats apply.

First, there is no reduction to a known computational hardness assumption. The contraction bound $\sigma_{\max} \approx 0.5$ was measured empirically at $n = 4$; while the TBJA attack at $n = 6$ is consistent with contraction, a formal proof that $\rho(D\Phi(x)) < 1$ holds uniformly over the reachable state space remains open.

Second, the parameter regime ($N = 32$, $T = 50$) is small enough that unconventional attacks exploiting the specific structure of Equation (2)—for example, algebraic attacks on the rational nonlinearity—cannot be ruled out without further analysis.

Third, floating-point determinism is platform-dependent. While WaveLock is deterministic under fixed hardware and software configurations, cross-platform reproducibility of ψ^* at the bit level requires careful control of FP64 rounding modes and operator ordering.

WaveLock is best interpreted as an *exploratory candidate* suggesting that nonlinear PDE evolution may generate one-way behavior of cryptographic interest.

8 Limitations

The principal limitations of this work are: the absence of a formal cryptographic security analysis; the unknown behavior of the operator under alternative discretizations, boundary conditions, or grid sizes; the limited understanding of stability bounds beyond the empirically measured Jacobian spectrum; potential sensitivity to the specific parameter values in Table 1; and the reliance on FP64 arithmetic, whose rounding behavior is implementation-defined beyond IEEE 754 guarantees.

9 Future Work

Formal analysis. Establish rigorous Lipschitz and contraction bounds on Φ as a function of the parameters $(\alpha, \beta, \theta, \varepsilon, \delta, \mu, \Delta t, T, N)$.

Brute-force evaluation for small N . Complete preimage enumeration is feasible for $n \leq 4$; this would provide exact measurements of preimage set sizes.

Alternative nonlinear operators. Evaluate other curvature feedback functions, including operators derived from the full Wave Confinement Theory master equation [15].

Scaling behavior. Study how the contraction rate, preimage entropy, and attack resistance scale with grid size N and iteration count T .

Protocol integration. Investigate the use of WaveLock commitments in machine-to-machine authentication protocols after formal security evaluation.

10 Conclusion

WaveLock demonstrates that deterministic nonlinear PDE evolution, when paired with structured serialization and hashing, can exhibit empirical one-way characteristics. The combined failures of gradient-based inversion, tangent-space recovery, reverse-time orbit construction, neural inverse mapping, and all tested classical and quantum-inspired heuristics are consistent with the strong contraction and information loss predicted by the Jacobian spectral analysis.

While no formal hardness claims are made, these results suggest that curvature-based cryptographic constructs—in which one-wayness emerges from the physics of contractive nonlinear evolution rather than from algebraic hardness assumptions—warrant further investigation as a complementary direction in cryptographic design.

A Non-Invertibility of the WaveLock Evolution Map

We formally derive consequences of the empirical Jacobian bound observed in the WaveLock evolution operator. Let $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ denote a single PDE update step, and let

$$\Phi_T = \underbrace{\Phi \circ \Phi \circ \dots \circ \Phi}_T$$

be the T -step evolution map so that $\psi^\star = \Phi_T(\psi_0)$. Let $J_1(x) = D\Phi(x)$ and $J_T(\psi_0) = D\Phi_T(\psi_0)$ denote the Jacobians.

Assumption A.1. For all x in the reachable state space, $\rho(J_1(x)) \leq \sigma < 1$.

Remark A.2. Assumption A.1 is supported by the empirical measurement $\sigma_{\max} \approx 0.5$ at $n = 4$ (Section 5.3). A formal proof that this bound holds uniformly for all reachable states and grid sizes remains open.

A.1 Contraction

Lemma A.3 (Contraction). *Under Assumption A.1, Φ is a contraction in the Euclidean norm:*

$$\|\Phi(x) - \Phi(y)\| \leq \sigma \|x - y\|.$$

Proof. By Gelfand's formula, $\rho(J_1) = \lim_{k \rightarrow \infty} \|J_1^k\|_2^{1/k}$. Since $\rho(J_1) < 1$, there exist $\sigma < 1$ and $C > 0$ such that $\|J_1^k\|_2 \leq C\sigma^k$. Linearizing, $\|\Phi(x + \delta) - \Phi(x)\| \leq \|J_1\|_2 \|\delta\| \leq \sigma \|\delta\|$. $\square$

Corollary A.4. *For all ψ_0 and perturbations δ ,*

$$\|\Phi_T(\psi_0 + \delta) - \Phi_T(\psi_0)\| \leq \sigma^T \|\delta\|.$$

A.2 Non-Injectivity

Theorem A.5 (No inverse exists). *If Φ is a strict contraction ($\sigma < 1$), then Φ_T is not injective. Consequently, Φ_T^{-1} does not exist as a function.*

Proof. For $x \neq y$, $\|\Phi_T(x) - \Phi_T(y)\| \leq \sigma^T \|x - y\|$. As T increases, $\sigma^T \|x - y\| \rightarrow 0$. In finite-precision arithmetic (FP64), any two values closer than machine epsilon $\approx 2.2 \times 10^{-16}$ coincide numerically. Thus there exist $x \neq y$ such that $\Phi_T(x) = \Phi_T(y)$, and Φ_T is not injective. $\square$

A.3 Exponential Preimage Sets

Let $H(\cdot)$ denote differential entropy. For differentiable maps,

$$H(\Phi_T(\psi_0)) = H(\psi_0) + \log |\det J_T|.$$

Lemma A.6. *If $\|J_1\|_2 \leq \sigma < 1$, then $|\det J_T| \leq \sigma^{Td}$.*

Proof. In the linearization $J_T \approx J_1^T$, we have $|\det J_T| \leq \|J_T\|_2^d \leq (\sigma^T)^d$. $\square$

Theorem A.7 (Exponential preimage). *If $\sigma < 1$, then the preimage of a typical output $\psi^\star$ under Φ_T contains on the order of $2^{Td|\log_2(\sigma)|}$ distinct inputs.*

Proof. From the entropy identity, $H(\Phi_T(\psi_0)) = H(\psi_0) + Td \log \sigma$. Since $\log \sigma < 0$, the output entropy is smaller than the input entropy by $Td|\log \sigma|$ bits. This information loss forces an exponential number of inputs to map to each output value. $\square$

A.4 Gradient Collapse Under Backpropagation

Lemma A.8 (Vanishing gradient). *For any loss $L(\psi^*)$,*

$$\|\nabla_{\psi_0} L\| = \|(J_T)^\top \nabla_{\psi^*} L\| \leq \sigma^T \|\nabla_{\psi^*} L\|.$$

Proof. Directly from $\|J_T\|_2 \leq \sigma^T$. Since $\sigma^T \ll 1$, gradients collapse. $\square$

A.5 Non-Existence of Reverse-Time Orbits

Theorem A.9 (Reverse evolution ill-posed). *If Φ is a contraction, then the reverse-time PDE $\psi_{t-1} = \Phi^{-1}(\psi_t)$ is ill-posed: no reverse trajectory terminating at ψ^* exists.*

Proof. Since Φ_T is not injective (Theorem A.5), Φ_T^{-1} cannot be defined on ψ^* . Any reverse-time evolution must implement Φ^{-1} , which does not exist as a function. Thus reverse trajectories do not exist in the sense of Hadamard [5]: no uniqueness, no stability, and no continuous dependence on data. $\square$

A.6 Universal Failure of Inversion Attacks

Theorem A.10 (Structural invertibility barrier). *Under Assumption A.1, every inversion method relying on gradients, linear structure, adjoint dynamics, learned inverse models, or reversible heuristics must fail.*

Proof. Each attack category is defeated by a specific consequence of contraction: gradient attacks fail because gradients collapse as σ^T (Lemma A.8); spectral, Fourier, and wavelet attacks fail because the spectrum is compressed by σ^T ; adjoint-PDE attacks fail because the backward PDE is ill-posed (Theorem A.9); neural inversion fails because the inverse map is not single-valued (Theorem A.5); and quantum-inspired attacks fail because amplitude amplification and spectral decomposition require approximate reversibility or sparse structure, both destroyed by contraction. $\square$

A.7 Consolidated Statement

Theorem A.11 (WaveLock non-invertibility and empirical one-wayness). *Let $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ satisfy Assumption A.1. Let Φ_T denote its T -fold composition and let $\psi^* = \Phi_T(\psi_0)$. Then:*

- (i) **Strict contraction.** $\|\Phi_T(x) - \Phi_T(y)\| \leq \sigma^T \|x - y\|$.
- (ii) **Non-injectivity.** For sufficiently large T , $\Phi_T(x) = \Phi_T(y)$ in finite precision for distinct x, y .
- (iii) **No inverse.** No single-valued map G satisfying $G(\psi^*) = \psi_0$ exists.
- (iv) **Exponential preimage sets.** The preimage of a typical ψ^* contains $\sim 2^{Td|\log_2(\sigma)|}$ distinct inputs.
- (v) **Gradient collapse.** $\|\nabla_{\psi_0} L\| \leq \sigma^T \|\nabla_{\psi^*} L\| \ll 1$.
- (vi) **Ill-posed backward dynamics.** The reverse-time PDE is ill-posed; no reverse orbit terminating at ψ^* exists.

(vii) **Universal failure of inversion methods.** Any adversary using gradients, spectral methods, adjoint PDEs, learned inverse models, or quantum-inspired reversible heuristics must fail.

Consequently, the composite map $G(\psi_0) = H(\text{Serialize}(\Phi_T(\psi_0)))$ behaves as an empirical one-way operator.

Proof. Items (i) and (ii) follow from Lemma A.3 and Corollary A.4 and the finite-precision argument. Item (iii) follows from non-injectivity. Item (iv) follows from Lemma A.6 and the entropy identity. Item (v) follows from $\|J_T\|_2 \leq \sigma^T$. Item (vi) follows from non-injectivity and Hadamard ill-posedness [5]. Item (vii) follows because every listed attack category requires non-vanishing gradients, approximate reversibility, a stable adjoint operator, or a learnable inverse manifold—all ruled out by strict contraction. $\square$

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